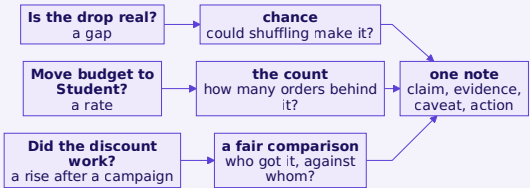


1 Three questions, three checks, one note



A gap gets a chance reference, a rate gets its count, and a campaign gets a fair comparison. Every check ends as one line of the same note.

Crux. A p-value is a share of chance-only worlds; it is never the chance the finding is wrong.

2 The shuffle test, in five moves

Move	What it does
The real gap	Q1 mean less Q2 mean, per member
Shuffle	Pool both quarters and deal the labels at random
Recompute	The gap in this chance-only world
Repeat	Thousands of times, with a fixed seed
Count	The share at least as large as the real gap

```
random.seed(2026)
for _ in range(5000):
    random.shuffle(pool)
    gaps.append(mean(pool[:k]) - mean(pool[k:]))
share = sum(1 for g in gaps if g >= real_gap) / len(gaps)
```

3 The sentence a share supports

Draft	Survives Kavya
"A 3 percent chance we are wrong."	"If nothing had changed, a gap this large turns up in about 3 of every 100 shuffles."
"97 percent certain."	"Chance rarely makes a gap this large, so we treat it as real."
"p = 0.34 proves nothing changed."	"The gap sits inside the usual wobble."
Counting falls only answers "a drop this big"; counting rises too roughly doubles the share.	

4 Real, then worth it

A small share says a gap beats chance, never that it is big: with enough orders even Rs 20 on a basket of about Rs 2,200 comes back significant. Size it three ways: per member, for the segment, against the company's quarter. Then set it against the cost of acting: a fix that costs C against a fall of F pays only if it wins back more than C over F.

Crux. Real and worth acting on are two separate calls: the shuffle answers the first, rupees against cost answer the second.

5 The count behind a rate

Orders behind the rate	One order moves it	Chance of a 40% rise from coin flips
10	10 points	about 38 in 100
30	3.3 points	about 18 in 100
100	1 point	about 4 in 100
400	0.25 points	under 1 in 1,000

Crux. Count what a rate stands on before you repeat it; under thirty, it is a lead.

6 Who got it, against whom

A blend can rise while every segment falls, when the group that got the campaign holds more of the segment that spends more anyway. Compare inside each segment, state the mix of the two groups, and settle it next time with a random held-back slice of each segment. A discount of d needs volume to rise by 1 / (1 minus d), less one, just to stand still: 17.6 percent at 15 percent off.

Crux. Split an aggregate by segment before you credit a campaign, and name who got it.

7 The note to Meera

Claim: one sentence she can act on. Evidence: the number with its count and how it was checked. Caveat: what would change the claim. Action: what to do next and what it costs, including "wait and measure".

Crux. The note is claim, evidence, caveat, action, and "not yet, and here is what would tell us" is a complete answer.

THE WORDS ON THIS SHEET

Label shuffle Dealing the quarter labels at random and recomputing the gap, to make a...	p-value The share of chance-only worlds with a gap at least as large as the real one	Lead A rate on fewer than thirty observations, watched and never acted on	Simpson's reversal A comparison that runs one way in every group and the other way in the blend	Held-back group A random slice inside each segment that does not get the campaign
Wobble The gap chance produces between two ordinary quarters	Confidence interval The range of sizes the data supports, named today and built later	Confounder A variable that drives both who got a campaign and the outcome		